

2020 2월 2일 대산고 (2월 2일) 기말 수능 (2월).

1. $f(x) = 2x + 3$

$f^{-1}(3) = a \rightarrow f(a) = 3$

$2a + 3 = 3$

$2a = 0$

$a = 0$

2. $g(ax+b)$

$= -(ax+b) + c$

$= -ax - b + c = x + 3$

$-a = 1$

$a = -1$

$-b + c = 3$

$f(1) = 1$

$a + b = 1$

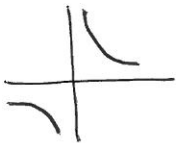
$b = 2$

$c = 5$

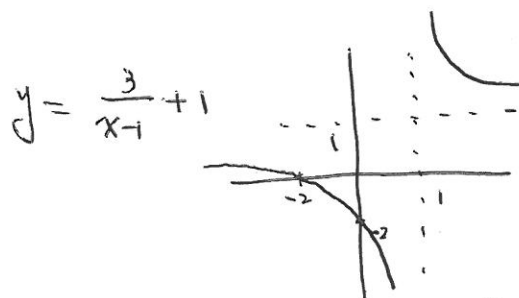
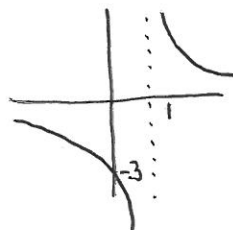
$a \cdot b \cdot c = -1 \times 2 \times 5$

$= -10$

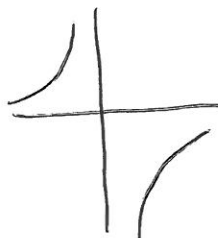
3.

$y = \frac{1}{x}$ 

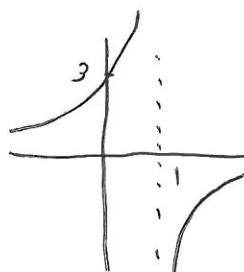
$y = \frac{3}{x-1}$



$y = \frac{-1}{x}$



$y = \frac{-3}{x-1}$



~~2020.2.2~~

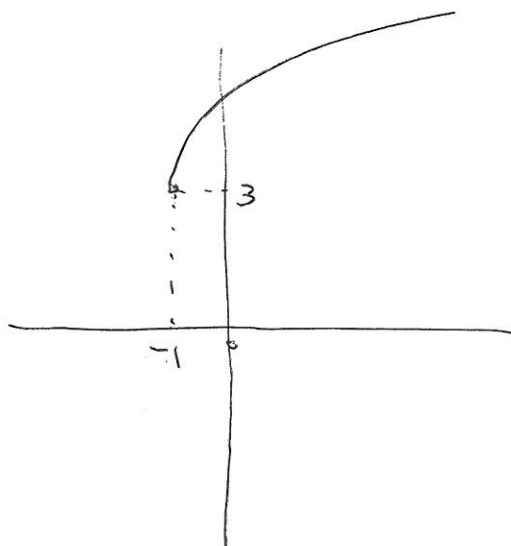
4. $f(x) = \sqrt{2x}$

$$x \rightarrow -1 \Rightarrow x = (x+1)$$

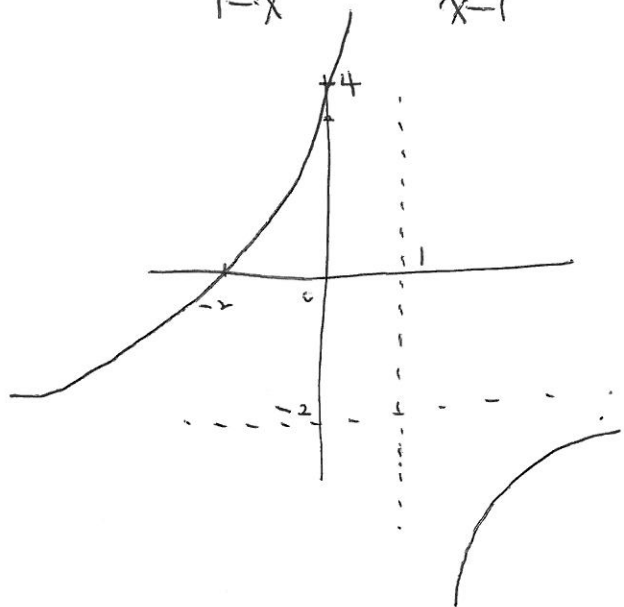
$$y \rightarrow 3 \Rightarrow y = (y-3)$$

$$y-3 = \sqrt{2(x+1)}$$

$$y = \sqrt{2(x+1)} + 3$$



5. $f(x) = \frac{4+2x}{1-x} = \frac{-6}{x-1} - 2$



$$6. {}^nC_4 = {}^nC_3 = \frac{n \times 6 \times 5}{3 \times 2 \times 1} = 35$$

7. $f: 4 \times 3 \times 2 \times 1 = 24$

$g: 4$

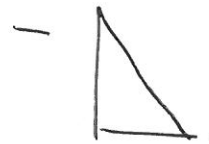
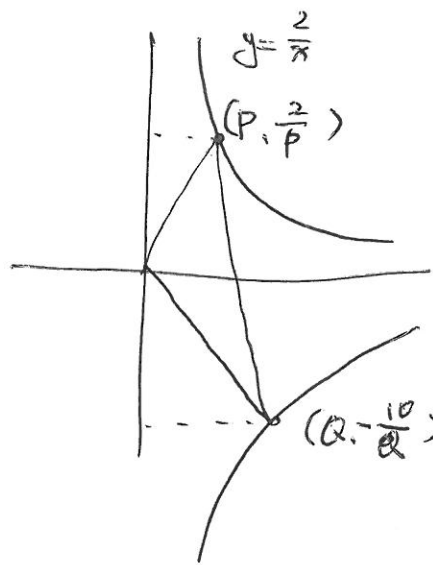
$h: 1$

$$a=24 \quad b=4 \quad c=1$$

$$a+b-c$$

$$= 24 + 4 - 1 = 27$$

8



$$\begin{aligned}
 &= (P+Q) \times \left(\frac{2}{P} + \frac{10}{Q} \right) \times \frac{1}{2} - P \times \frac{2}{P} \times \frac{1}{2} - Q \times \frac{10}{Q} \times \frac{1}{2} \\
 &= \left(\frac{10P}{Q} + \frac{2Q}{P} + 12 \right) \times \frac{1}{2} - 1 - 5 \\
 &= \left(\frac{10P}{Q} + \frac{2Q}{P} + 12 \right) \times \frac{1}{2} - 6 = \frac{5P}{Q} + \frac{Q}{P}
 \end{aligned}$$

$$\frac{5P}{Q} + \frac{Q}{P} \geq 2\sqrt{5}$$

Q. 1. $\oplus 2C_1$. 2. 4. 6. 8. 10. 12. 14. 16 ~~16~~ ~~8~~ $8C_2$

$\oplus 3C_1$ 3. 6. 9. 12. 15 $5C_2$.

$\oplus 5C_1$ 5. 10. 15 $3C_2$

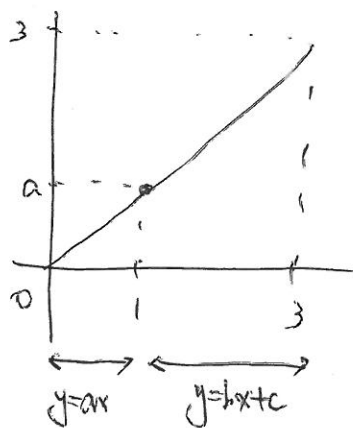
$\ominus 6C_1$ 6. 12 $2C_2$

$\oplus 1C_1$ 1. 14 $2C_2$

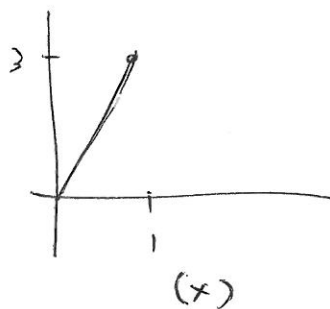
$$8C_2 + 5C_2 + 3C_2 - 2C_2 + 2C_2$$

$$= 28 + 10 + 3 - 1 + 1 = 41.$$

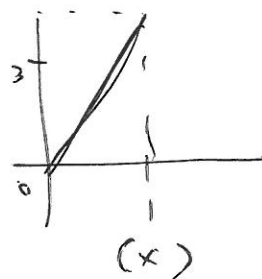
10.



7. $a=3$



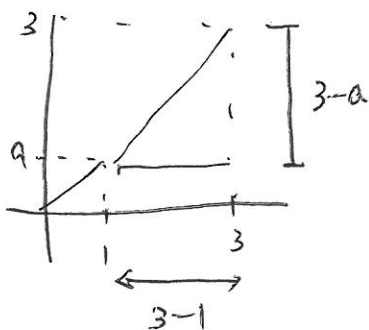
$a > 3$



$0 < a < 3$

L. $ab > 0$

$a > 0, b > 0$



$\frac{3-a}{2} = b$

$3-a=2b$

$a+2b=3$

L. $ab < 0, a > 0, b < 0$ (x)

11. $y = \frac{ax+b}{x+c} = \frac{b-ac}{x+c} + a \rightarrow y = \frac{4x+b}{x+1} \Rightarrow y = \frac{4x-8}{x+1}$

$A = \{x | x \neq -c, \forall c\}$

$B = \{y | y \neq a, \forall c\}$

$A-B = +a = 4$

$B-a = -c = -1$
 $c = 1$

$a=4, b=-8, c=1$

$a+b+c = 4-8+1 = -3$

12.

A-4

B-3

C-2

D-3

E-2

$$4 \times 3 \times 2 \times 3 \times 2 = 144$$

13. 2nd case $6C_3 \times 3C_3 \times 3! \times \frac{1}{2} = 120$
 $\oplus$ 420.

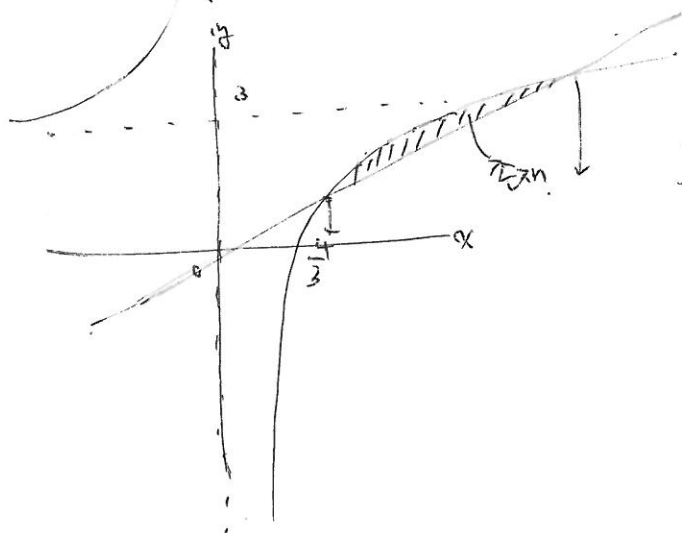
3rd case $6 \times 5C_3 \times 2C_2 \times 3! = 360$

14. $y = \frac{3x-4}{x}$
 $= \frac{-4}{x} + 3$

$$y = \sqrt{3a-x} + a$$

$$= \sqrt{-(x-3a)} + a.$$

$$(3a, a) \rightarrow y = \frac{1}{3}x \text{ at } x=9$$



$$\frac{3x-4}{x} = \frac{1}{3}x$$

$$9x-12=x^2$$

$$x^2-9x+12=0 \rightarrow \alpha, \beta$$

$$\alpha \leq 3a \leq \beta$$

$$\frac{\alpha}{3} \leq a \leq \frac{\beta}{3}$$

$$M = \frac{\beta}{3} \quad m = \frac{\alpha}{3}$$

$$\frac{1}{3}(\alpha+\beta) = \frac{1}{3} \times 9 = 3.$$

15.

$$100 < \boxed{abc} < 1060$$

$$A = 1, 2, 3, 4, 5, 6$$

$$C = 1, 3, 5, 7, 9$$

$$a \wedge c = 1, 3, 5 \dots$$

$$a \rightarrow 1, 3, 5 \quad c \rightarrow 4 \times 2 \quad b \rightarrow 8 \times 2$$

$$3 \times 4 \times 8 = 96$$

$$b \rightarrow 2, 4, 6 \quad c \rightarrow 5 \times 2 \quad b \rightarrow 8 \times 2$$

$$3 \times 5 \times 8 = 120 \times 2$$

$$a \rightarrow 7 \quad c \rightarrow 1, 3, 5 \quad b \rightarrow 5 \times 2$$

$$\cancel{3 \times 5} = 15$$

$$a \rightarrow 7 \quad c \rightarrow 9 \quad b \rightarrow 6 \times 2$$

$$96 + 120 + 15 + 6 = 237$$

답답해 1.

$$f(2) < f(4) < f(6) < f(8)$$

$$8C_4 \times 4P_3 = 1680 > 1$$

답답해 2.

$$f(x) = ax + b$$

$$f(1) = a + b = 3$$

$$f(2x) = 2ax + b \quad 2f(x) = 2(ax + b) \\ = 2ax + 2b$$

$$b = 2b \\ b = 0$$

$$f(x) = 3x$$

$$g(x) = \frac{3x - 9}{3x + 9} = \frac{x - 3}{x + 3} = \frac{(x + 3) - 6}{x + 3} = \frac{-6}{x + 3} + 1$$

$$P\left(t, \frac{-6}{t+3} + 1\right) \quad A(-3, 1)$$

$$|xy| = \sqrt{\left(\frac{t}{t+3}\right)^2 + \left(\frac{-6}{t+3}\right)^2}$$

$$= \sqrt{\left(\frac{t}{t+3}\right)^2 + \frac{36}{(t+3)^2}}$$

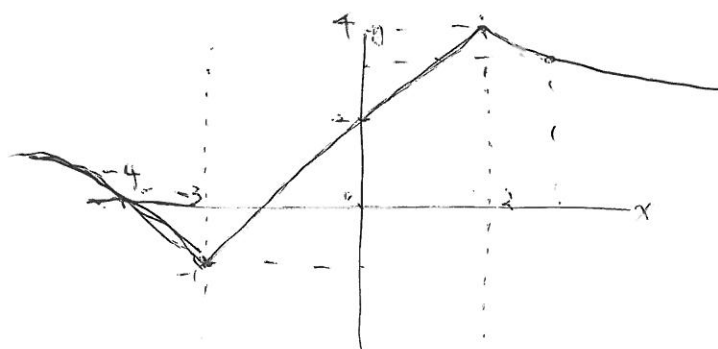
$$= \sqrt{12}$$

$$= 2\sqrt{3}$$

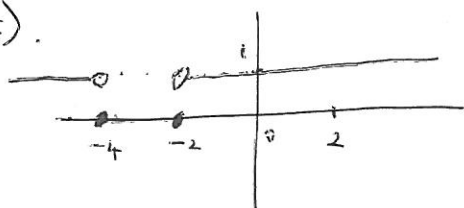
$$\begin{aligned} \left(\frac{t}{t+3}\right)^2 + \frac{36}{(t+3)^2} &\geq 2\sqrt{36} \\ &\geq 12 \end{aligned}$$

답답해 3.

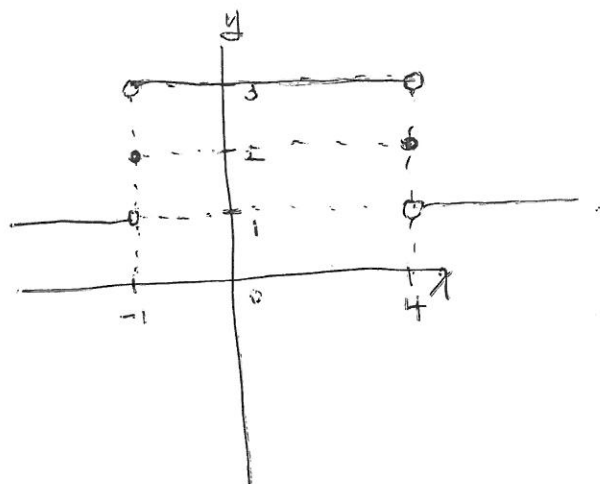
$$f(x) = \begin{cases} \sqrt{-(x+3)} - 1 & (x \leq -3) \\ x+2 & (-3 < x < 2) \\ -\sqrt{x-2} + 4 & (x \geq 2) \end{cases}$$



$g(t)$.



$h(t)$



$$g(t) + h(t) = 2 \quad (t < -4)$$

$$1 \quad (-4 \leq t \leq -2)$$

$$2 \quad (-2 < t < -1)$$

$$3 \quad (t = -1)$$

$$\underline{4 \quad (-1 < t < 4)}$$

$$3 \quad (t = 4)$$

$$2 \quad (t > 4)$$

$$t = 0, 1, 2, 3$$

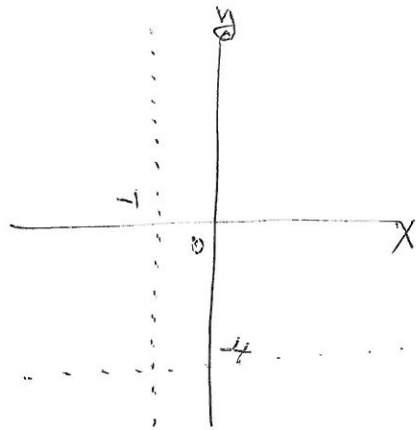
⑥

1. ~~1. 2. 3. 4.~~

$$y = \frac{-4x + k^2 - 2k - 7}{x+1}$$

$$= \frac{-4(x+1) + 4 + k^2 - 2k - 7}{x+1}$$

$$= \frac{k^2 - 2k + 3}{x+1} - 4.$$



1) 3. 4. ~~1. 2.~~ $k^2 - 2k + 3 = 0.$

$$(k+1)(k-3) = 0$$

$$\underline{k = -1 \text{ or } 3.}$$

2) 2. 3. 4. ~~1. 5.~~

~~$$k^2 - 2k + 3 > 0 \quad (0, 0) \downarrow$$~~

~~$$k < -1, k > 3$$~~

$$k^2 - 2k + 3 < 0.$$

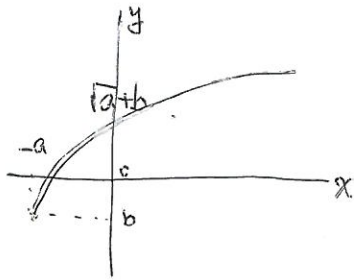
$$\underline{-1 < k < 3.}$$

3) ~~1. 2. 3. 4.~~

$$k^2 - 2k + 3 > 0.$$

$$\underline{k < -1 \text{ or } k > 3}$$

$$A+B=12$$

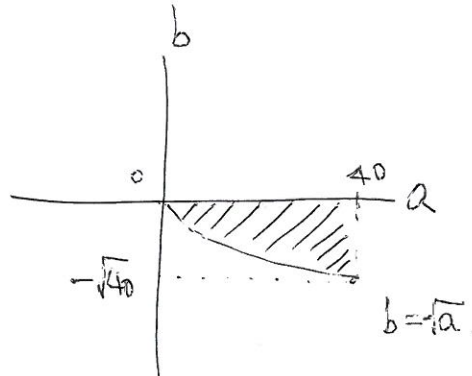


$$-a < 0$$

$$a > 0 \quad b < 0$$

$$\sqrt{a+b} > 0 \rightarrow b > -\sqrt{a}$$

$$a=1 \quad b > -1 \quad b = x$$



$$a=2, 3, 4 \rightarrow b=-1 \quad 3 \times 1 = 3 > 1$$

$$a=5 \dots 9 \rightarrow b=-1, -2, \dots \quad 5 \times 2 = 10 > 1$$

$$a=10 \sim 16 \rightarrow b=-1, -2, -3 \quad 11 \times 3 = 21 > 1$$

$$a=17 \sim 25 \rightarrow b=-1, -2, -3, -4 \quad 9 \times 4 = 36 > 1$$

$$a=26 \sim 36 \rightarrow b=-1, -2, -3, -4, -5 \quad 11 \times 5 = 55 > 1$$

$$a=37 \sim 40 \rightarrow b=-1, -2, -3, -4, -5, -6 \quad 4 \times 6 = 24 > 1$$

$$\underline{149 > 1}$$